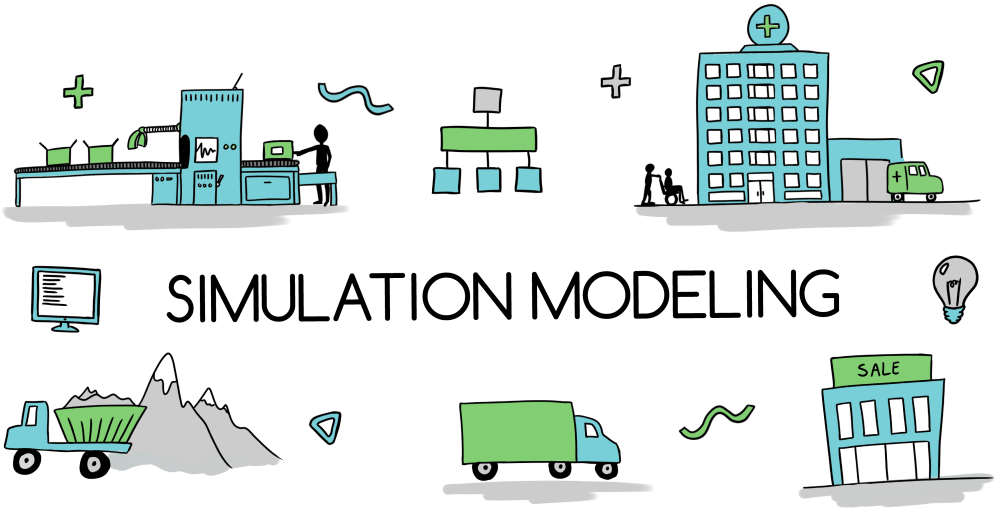
IE 7215: Simulation Analysis



Prof. Mohammad Dehghani

"Logic is the foundation of the certainty of all the knowledge we acquire.""
- Leonhard Eule

Simulation Analysis

IE7215

Instructor Info

Mohammad Dehghani

Office Hrs:
(Make an appointment)

Website

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Course Info

Prerequisite:
* IE 6200 - Engineering Probability and Statistics.
* Computing, particularly use of Excel and basic programming.

Online: Asynchronous

TA Info

Check Canvas

Check Canvas

Zoom Meeting

Check Canvas

Course Overview

This is an introductory course to simulation, a numerical approach that uses computers to model complex stochastic systems that are difficult to understand via analytical methods. We will start with foundations of random variables generation for both discrete and continuous cases. General principles of spreadsheet and systems simulation will be covered, using @Risk (www.palisade.com) and Simio (www.simio.com) software for class projects. The second part of the course focuses on statistical analysis and validation techniques. Upon completion of the course students will be able to develop complex simulation models, design the simulation experiment to be run on the model, and analyze and interpret the results. Along with it, management of simulation projects is also addressed.

Course Objectives

- Students will learn system theory and basic concepts in discrete-event simulation (DES) field: modeling, data collection for input parameters, output analysis, verification and validation, and experimentation
- Students will learn creating a simulation model using software (Simio) to improve or design an industrial and a business system
- Students will develop critical thinking and creativity skills through project and design of experiment (sensitivity analysis)
- Student's communication skills will improve through writing reports, teamwork, and occasional presentations

Textbook

- Main Textbook
  - Simio and Simulation: Modeling, Analysis, Applications - New 4th Edition by Jeffrey S. Smith, David T. Sturrock (Link)
  - This book is required for the class (either the hard-copy or its PDF version). Along with that, several articles will be provided.
  - Edition Notice: The course contents are based on the 4th Edition of the textbook, while the new online version corresponds to the 7th Edition. Although there are updates in the newer edition, the foundational concepts remain relevant and applicable to the course.

Software Tools

- Software Packages
  - @RISK: it's trial version is free, and would be sufficient for our class
  - Simio: Detailed instruction on how to access this software can be found in Course Setup Instruction file uploaded on CANVAS.

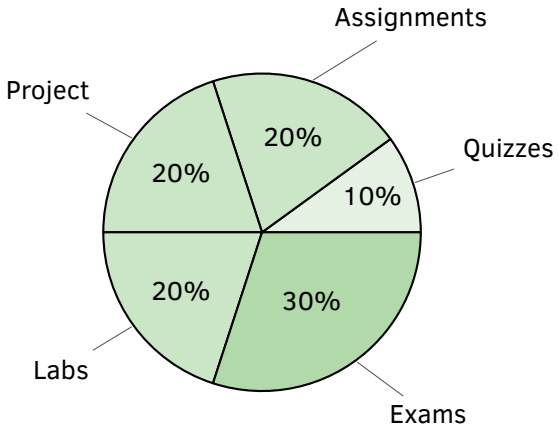
Course Schedule

Please check Course Schedule file posted on Canvas for the weekly schedule of the course.

Outline

- Monte-Carlo Simulation (MCS)
  - Probability Distribution Functions
  - Random Number Generation (RNG)
  - Distribution-fitting and Input Modeling
- Queuing Theory
  - Queuing and Queuing Networks
  - Modeling and Debugging
- Discrete Event Simulation (DES)
  - Fundamental Simulation Concepts
  - Spreadsheet Simulation Overview
  - Simulation modeling in Simio
  - Simulation Output Analysis
  - Optimization via Simulation

Grading Scheme



- Assignments and Labs (Individual)
  - Assignments will be released on a weekly basis.
  - All homework shall be submitted through appropriate mediums as communicated by the professor/TAs
  - If due to unforeseen circumstances the schedule changes, students will be informed prior to the change in schedule. Submissions have to be labeled as "IE 7215 - HW # (or Lab #) - Sec 2 - <Student Name>"
- Quizzes (Individual)
  - Since the concepts in this course are interrelated and often build off each other, you will be assessed on your cumulative understanding of course topics. The quizzes will be posted online through CANVAS and you have just one attempt to take each quiz
  - Once a quiz is posted, you have a few days (usually 5 days) to take it. However, the quiz duration is limited and you have to save and submit your answers within the given time. Since the quizzed are online, the missed quizzed will not be made up.
- Exams (Individual)
  - 3 exams: Midterm, Final Exam (Theory and Practical) will be posted on CANVAS according to the schedule.
  - Final Exam is a combination of theory and practicals. The practical part of this exam will be based on simulation modeling and analysis in Simio. More details will be posted in CANVAS.
  - will be based on simulation model creation and analysis in Simio. This exam will be given to students at the end of the semester.
- Project (Group based)
  - The projects provide an opportunity to apply core principles of Simulation to solve your preferred complex and optimization problem.
  - The project can be done individually or in a group of 2 students.
  - Please refer to the suggested project timeline to make sure that you have sufficient time or experimentation and analysis.
  - Project has two phases: *(1) Proposal Presentation*, and *(2) Final Presentation*.The details about the project will be announced.
- Class Participation & Discussion
  - The course is synchronous and attendance is mandatory for students taking the course online or on-ground. To have an effective learning experience, students are encouraged to have continuous interaction with the faculty. A discussion question will be given to each section/break-out rooms to stimulate critical thinking. All students need to actively participate in discussions to articulate their ideas and share their thoughts.
  - Performance will be evaluated based on their participation and quality of response.

Grading

Your grades will be posted on CANVAS. You can check your running grade at any time, so there are no surprises. In fact, you should audit the correct entry of your grades. No grade corrections will be made after the last week of the semester. Final Grades and Course Letter Grades are not curved and are assigned as follows:

Letter Grade	Points	Letter Grade	Points	Letter Grade	Points
A	93-100	B	83-86	C	73-76
A-	90-92	B-	80-82	C-	70-72
B+	87-89	C+	77-79	F	<70

Emailing Us?

Please use the following format for the subject line of your email to avoid miss communication. This subject line will reduce the confusion.

IE7215 - Sec # - Subject Line

Make-up Policy

- No makeup opportunity will be given for any homeworks/lab assignment/projects.
- Students unable to attend classes are responsible for obtaining information regarding announcements and lecture content

Class Policy

- Students are expected to attend all class sessions and are responsible for all materials presented in class. Past experience has shown students who attend regularly and interact with instructors/classmates tend to be better overall.
- Laptops: You may need to use your laptop during the class upon the instructor's notes
- Phones/tablets/smart watches: Electronic devices may not be used during the class. If you have class work, social networking or sports-score browsing, that is more important to do, then please skip class. Life is full of such choices.
- Discussion between students for homeworks and projects is permissible. However the ideas / solutions / codes must not be copied. Each individual submission must be an outcome of individual effort

Recording of Classes

Classes will be recorded to enable all students to review material covered in synchronous classes. Please contact me if you have any concerns.

Academic Integrity

A commitment to the principles of academic integrity is essential to the mission of Northeastern University. The promotion of independent and original scholarship ensures that students derive the most from their educational experience and their pursuit of knowledge. Academic dishonesty violates the most fundamental values of an intellectual community and undermines the achievements of the entire University.

As members of the academic community, students must become familiar with their rights and responsibilities. In each course, they are responsible for knowing the requirements and restrictions regarding research and writing, examinations of whatever kind, collaborative work, the use of study aids, the appropriateness of assistance, and other issues. Students are responsible for learning the conventions of documentation and acknowledgment of sources in their fields. Northeastern University expects students to complete all examinations, tests, papers, creative projects, and assignments of any kind according to the highest ethical standards, as set forth either explicitly or implicitly in this Code or by the direction of instructors.

Go to Academic Integrity website to access the full academic integrity policy.

Student accommodations

Northeastern University and the Disability Resource Center (DRC) are committed to providing disability services that enable students who qualify under Section 504 of the Rehabilitation Act and the Americans with Disabilities Act Amendments Act (ADAAA) to participate fully in the activities of the university. To receive accommodations through the DRC, students must provide appropriate documentation that demonstrates a current substantially limiting disability.

For more information, visit Disability Resource Center website

Northeastern University Copyright Statement

This course material is copyrighted, and all rights are reserved by Northeastern University. No part of this course material may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any language or computer language, in any form or by any means, electronic, mechanical, magnetic, optical, chemical, manual, or otherwise, with the express prior written permission of the University.

Diversity and Inclusivity Statement

Northeastern University is committed to equal opportunity, affirmative action, diversity and social justice while building a climate of inclusion on and beyond campus. In the classroom, members of the University community work to cultivate an inclusive environment that denounces discrimination through innovation, collaboration and an awareness of global perspectives on social justice. It is my intention that students from all backgrounds and perspectives will be well served by this course, and

that the diversity that students bring to this class will be viewed as an asset. I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, socioeconomic background, family education level, ability - and other visible and nonvisible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. Your suggestions are encouraged and appreciated.

Please visit the website for complete information on Diversity and Inclusion.

TITLE IX

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance.
Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including male, female, transgender students, faculty and staff.
In case of an emergency, please call 911.

Please visit the website for a complete list of reporting options and resources both on- and off-campus

Hybrid NUflex

Hybrid NUflex allows some portion of the students in each class to be remote and others to be in class. If you have the opportunity to attend class in-person, please come as often as itâs possible for you to do so. Remember your commitment to your faculty and classmates:

- Use the dynamic scheduling tool to set your preferences for in-person or remote attendance. Each week between Monday and Wednesday, you' ll select whether you prefer to attend your class sessions remotely or in-person for the following week. You will be able to choose a preference for each individual class session. For example, you can request to come one of your courses in-person on Monday and Wednesday and remotely on Friday.
- Follow through when you commit to attend class in-person. Space in the classroom is limited, and your fellow students and faculty want to see you.
- If you change your mind about in-person class attendance, update your preferences in the dynamic scheduling tool so someone else can take your seat. The dynamic scheduling tool creates a wait-list for students who would like to attend class in-person but received a remote assignment. Beginning in late January or early February, if youâre assigned an in-person seat in a class session for the following week, you have until Saturday at 11:59pm ET to switch to remote for that class session. Switching from in-person to remote frees up a seat for your fellow students on the wait-list to come to class.